1. Create the LOC_TEST table with the following structure:

```
CREATE TABLE LOC_TEST(
  location_id number(6) PRIMARY KEY,
  street varchar2(40),
  postal_code char(40),
  city char(40),
  state char(40)
)
```

Column Name	Data Type	Nullable	Default	Primary Key
LOCATION_ID	NUMBER(6,0)	No	-	1
STREET	VARCHAR2(40)	Yes	-	-
POSTAL_CODE	CHAR(40)	Yes	-	-
CITY	CHAR(40)	Yes	-	-
STATE	CHAR(40)	Yes	-	-

2. Create the DEPT_TEST table with the following structure:

```
CREATE TABLE DEPT_TEST(
  department_id number(6) PRIMARY KEY,
  department_name char(20),
  location_id number(6)
)
```

Column Name	Data Type	Nullable	Default	Primary Key
DEPARTMENT_ID	NUMBER(6,0)	No	-	1
DEPARTMENT_NAME	CHAR(20)	Yes	-	-
LOCATION_ID	NUMBER(6,0)	Yes	-	-

3. Create the EMPL_TEST table with the following structure:

```
CREATE TABLE EMPL_TEST(
  employee_id number(8) PRIMARY KEY,
  first_name char(30),
  last_name char(40),
  hire_date date,
  salary number(8,2),
  department_id number(6)
)
```

Column Name	Data Type	Nullable	Default	Primary Key
EMPLOYEE_ID	NUMBER(8,0)	No	-	1
FIRST_NAME	CHAR(30)	Yes	-	-
LAST_NAME	CHAR(40)	Yes	-	-
HIRE_DATE	DATE	Yes	-	-
SALARY	NUMBER(8,2)	Yes	-	-
DEPARTMENT_ID	NUMBER(6,0)	Yes	-	-

4. Insert rows into the tables created based on the following tables:

LOC_TEST

LOCATION_ID	STREET	POSTAL_CODE	CITY	STATE
1400	2014 Jabberwocky Rd	26192	Southlake	Texas
1500	2011 Interiors Blvd	99236	South San Francisco	California
1700	2004 Charade Rd	98199	Seattle	Washington
1800	460 Bloor St	On M551X8	Toronto	Ontario
2500	Magdalene Center	OX9 9ZB	Oxford	Oxford

```
insert all into LOC_TEST
values (1400, '2014 Jabberwocky Rd', 26192, 'Southlake', 'Texas')
INTO LOC_TEST VALUES (1500, '2011 Interiors Blvd', 99236, 'South San Francisco',
'California')
INTO LOC_TEST VALUES (1700, '2004 Charade Rd', 98199, 'Seattle', 'Washington')
INTO LOC_TEST VALUES (1800, '460 Bloor St', 'On M551X8', 'Toronto', 'Ontario')
INTO LOC_TEST VALUES (2500, 'Magdalene Center', 'OX9 9ZB', 'Oxford', 'Oxford')
select 1 from dual;
```

DEPT_TEST

DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID
60	IT	1400
50	Shipping	1500
80	Sales	2500

```
insert all into DEPT_TEST
values (60, 'IT', 1400)
INTO DEPT_TEST VALUES (50, 'Shipping', 1500)
INTO DEPT_TEST VALUES (80, 'Sales', 2500)
select 1 from dual;
```

EMPL_TEST

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	SALARY	DEPARTMENT_ID
141	Trenna	Rajs	17-Oct-1995	6945.23	50
143	Randall	Matos	15-Mar-1998	5674.68	50
174	Ellen	Abel	11-May-1996	7894.38	80
176	Jonathan	Taylor	24-Mar-1998	6784	80
144	Peter	Vargas	09-Jul-1998	4578	50
142	Curtis	Davies	29-Jan-1997	5684.9	50
149	Eleni	Zlotkey	29-Jan-2000	7956.35	80
124	Kevin	Mourgos	16-Nov-1999	6985.25	50
103	Alexander	Hunold	03-Jan-1990	8756	60
104	Bruce	Ernst	21-May-1991	6534.72	60
107	Diana	Lorentz	07-Feb-1999	5342.82	60

```
insert all into EMPL_TEST
values (141, 'Trenna', 'Rajs', '17-Oct-1995', '6945.23', 50)
INTO EMPL_TEST VALUES (143, 'Randall', 'Matos', '15-Mar-1998', '5674.68', 50)
INTO EMPL_TEST VALUES (174, 'Ellen', 'Abel', '11-May-1996', '7894.38', 80)
INTO EMPL_TEST VALUES (176, 'Jonathan', 'Taylor', '24-Mar-1998', '6784', 80)
INTO EMPL_TEST VALUES (144, 'Peter', 'Vargas', '09-Jul-1998', '4578', 50)
INTO EMPL_TEST VALUES (142, 'Curtis', 'Davies', '29-Jan-1997', '5684.9', 50)
INTO EMPL_TEST VALUES (149, 'Eleni', 'Zlotkey', '29-Jan-2000', '7956.35', 80)
INTO EMPL_TEST VALUES (124, 'Kevin', 'Mourgos', '16-Nov-1999', '6985.25', 50)
INTO EMPL_TEST VALUES (103, 'Alexander', 'Hunold', '03-Jan-1990', '8756', 60)
INTO EMPL_TEST VALUES (104, 'Bruce', 'Ernst', '21-May-1991', '6534.72', 60)
INTO EMPL_TEST VALUES (107, 'Diana', 'Lorentz', '07-Feb-1999', '5342.82', 60)
select 1 from dual;
```

Use EMPL_TEST, DEPT_TEST and LOC_TEST for the following:

1.
Arrange all the employees using hire_date and display the first_name, the last_name and the salary.

SELECT first_name, last_name, salary
FROM EMPL_TEST
ORDER BY hire_date;

2.
Display the minimum salary for employees having department_id=60 or 50.

SELECT MIN(salary)
FROM EMPL_TEST
WHERE department_id IN (60, 50);

3.
Display the first_name and the last_name in a single column for employees having department_id=50.

SELECT first_name ||' ' || last_name
FROM EMPL_TEST
WHERE department_id = 50;

4.
Display the first hired employee having department_id = 50.

SELECT *
FROM EMPL_TEST
WHERE hire_date = (SELECT min(hire_date) FROM EMPL_TEST WHERE department_id = 50);

5.
Increase the salary of employees having department_id=50 by 200 euros and display the salary, the last_name, the first_name and the new salary.

SELECT salary, last_name, first_name, salary+200.
FROM EMPL_TEST
WHERE department_id = 50;

6.
Display the average salary for employees where department_id = 50 or 80 and round to two decimal places.

SELECT ROUND(AVG(salary), 2)
FROM EMPL_TEST
WHERE department_id IN (50, 80);

7.
Arrange in descending order according to last_name and display first_name, last_name, salary and hire_date for employees having department_id=80 and salary >6500.

```
SELECT first_name, last_name, salary, hire_date
FROM EMPL_TEST
WHERE department_id=80 and salary >6500
ORDER BY last_name DESC;
```

8.
Increase the salary by 5% and display the first_name, the last_name, the salary, and the new salary.

```
SELECT first_name, last_name, salary, salary+salary*0.05
FROM EMPL_TEST;
```

9.
Using a substitution variable for department_id, display the maximum salary for each department.

```
SELECT MAX(salary)
FROM EMPL_TEST
WHERE department_id=:enter_dept_id;
```

10.
Display the latest hire date for department_id=80.

```
SELECT *
FROM EMPL_TEST
WHERE hire_date = (SELECT max(hire_date) FROM EMPL_TEST WHERE department_id = 80);
```

11.
Display the last in the alphabetic list of names name for department_id=50.

```
SELECT first_name
FROM EMPL_TEST
WHERE department_id = 50
ORDER BY first_name DESC;
```

12.
Display the last_name, first_name and the salary for employees havind the salary greater than the average of salaries for employees having department_id=50.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT AVG(salary) FROM EMPL_TEST WHERE department_id=50);
```

13.
Display the first_name, the last_name, and the salary for employees hired before Rajs.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE hire_date < (SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Rajs');
```

14.
Display the first_name, the last_name, and the salary for employees who earn less than the average salaries for all the employees having department_id=50 or 80.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary < (SELECT AVG(salary) FROM EMPL_TEST WHERE department_id in (50,
80));
```

15.
Display first_name, last_name, and the salary from employees who earn more than the minimum salaries for employees hired after Vargas.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT min(salary) FROM EMPL_TEST WHERE hire_date <
(SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Vargas'
));
```

16.
Display the first_name, the last_name, and the salary for employees hired before Vargas.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE hire_date < (SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Vargas');
```

17.
Display the first_name, the last_name, and the salary for employees who earn less than Hunold.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary < (SELECT salary FROM EMPL_TEST WHERE last_name = 'Hunold');
```

18.
Display the first_name, the last_name, and the salary for employees having the same department_id as Matos.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE department_id = (SELECT department_id FROM EMPL_TEST WHERE last_name =
'Matos');
```

19.
Display all those employees who earn more than Lorentz and are in the same department as Zlotkey.
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT salary FROM EMPL_TEST WHERE last_name = 'Lorentz') and department_id =
(SELECT department_id FROM EMPL_TEST WHERE last_name = 'Zlotkey')
;

20.
Display all those employees who have the same department_id as Rajs and were hired before Zlotkey.
SELECT *
FROM EMPL_TEST
WHERE hire_date < (SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Zlotkey') and department_id =
(SELECT department_id FROM EMPL_TEST WHERE last_name = 'Rajs');

21.
Arrange all the employees having the same department as Zlotkey using salary, and display first_name, last_name and the salary.
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE department_id = (SELECT department_id FROM EMPL_TEST WHERE last_name = 'Zlotkey');

22.
Increase the salary by 500 and display the first_name, the last_name, the salary, and the new salary for employees hired after Zlotkey.
SELECT first_name, last_name, salary, salary+500
FROM EMPL_TEST
WHERE hire_date > (SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Zlotkey');

23.
Display first_name, last_name, and the salary from employees who earn more than the minimum salaries for employees havind department_id=50, 80 or 90.
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT min(salary) FROM EMPL_TEST WHERE department_id in (50, 80, 90));

24.
Display the first_name, the last_name, and the salary for employees who earn more than the maximum salaries for all the employees having department_id=50.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT MAX(salary) FROM EMPL_TEST WHERE department_id = 50);
```

25.
Display first_name, last_name, and the salary from employees who earn more than the minimum salaries for employees hired after Matos.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT MIN(salary) FROM EMPL_TEST WHERE hire_date >
(SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Matos'));
```

26.
Display the first_name, the last_name, and the salary for employees hired before Taylor.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE hire_date < (SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Taylor');
```

27.
Display all the information about employees having last_name starting with „M”.

```
SELECT *
FROM EMPL_TEST
WHERE last_name LIKE 'M%';
```

28.
Display first_name with capital letter, the last_name and the salary for employees having department_id =50 or 60

```
SELECT UPPER(first_name), first_name, last_name, salary
FROM EMPL_TEST
WHERE department_id IN (50, 60);
```

29.
Display first_name, last_name, and the salary from employees who earn more than the minimum salaries for employees hired after Davies.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT MIN(salary) FROM EMPL_TEST WHERE hire_date >
(SELECT hire_date FROM EMPL_TEST WHERE last_name = 'Davies'));
```

30.
Create a join between the tables EMPL_TEST and DEPT_TEST and display all the information from both tables for employees having department_id=80.

```
SELECT *
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE department_id=80;
```

31.
Display the first_name, the last_name, and the salary for employees who earns more than the maximum salaries for all the employees having department_id=60.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT MAX(salary) FROM EMPL_TEST WHERE department_id = 60);
```

32.
Display the first_name, the last_name, and the salary for employees who earns more than Rajs.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST
WHERE salary > (SELECT salary FROM EMPL_TEST WHERE last_name = 'Rajs');
```

33.
Display the first_name, the last_name, and the salary for employees having department_name=IT

```
SELECT first_name, last_name, salary
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE department_name='IT';
```

34.
Display the first_name, the last_name, and the salary for employees having location_id=1500.

```
SELECT first_name, last_name, salary
FROM EMPL_TEST NATURAL JOIN LOC_TEST
WHERE location_id=1500;
```

35.
Display the first_name, the last_name, the salary, the department_name and the location_id for employees having the same department_id as Rajs.

```
SELECT first_name, last_name, salary, department_name, location_id
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE department_id = (SELECT department_id FROM EMPL_TEST WHERE last_name =
'Rajs');
```

36.
Display first_name, last_name, the salary, the department_name, and the location_id for employees who earn more than Mourgos.

```
SELECT first_name, last_name, salary, department_name, location_id
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE salary > (SELECT salary FROM EMPL_TEST WHERE last_name = 'Mourgos');
```

37.
Create a join between EMPL_TEST and DEPT_TEST using department_id and display all the information about employees who doesn't work in the same department as Rajs. Arrange all the employees using the last_name.

```
SELECT *
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE department_id != (SELECT department_id FROM EMPL_TEST WHERE last_name =
'Rajs')
ORDER BY last_name;
```

38.
Create a join between the tables EMPL_TEST and DEPT_TEST and display all the information from both tables for employees having department_id=80 or 50.

```
SELECT *
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE department_id IN (80, 50);
```

39.
Create a join between the tables LOC_TEST and DEPT_TEST using location_id and display all the information from both tables.

```
SELECT *
FROM LOC_TEST JOIN DEPT_TEST USING(location_id);
```

40.
Create a join between the tables EMPL_TEST and DEPT_TEST using department_id and display the first_name, lat_name, department_name and location_id.

```
SELECT first_name, last_name, department_name, location_id
FROM EMPL_TEST JOIN DEPT_TEST USING(department_id);
```

41.
Create a join between the tables EMPL_TEST, LOC_TEST and DEPT_TEST and display all the information for employees having department_id=80.

```
SELECT *
FROM EMPL_TEST NATURAL JOIN DEPT_TEST NATURAL JOIN LOC_TEST
WHERE department_id = 80;
```

42.
Create a join between EMPL_TEST and DEPT_TEST using department_id and display all the information about employees having department_name=IT.

```
SELECT *
FROM EMPL_TEST NATURAL JOIN DEPT_TEST
WHERE department_name = 'IT';
```

43.
Display the first_name, last_name, the street, the city and the state for employees having the same department_id as Mourgos.

```
SELECT first_name, last_name, street, city, state
FROM EMPL_TEST NATURAL JOIN DEPT_TEST NATURAL JOIN LOC_TEST
WHERE department_id = (SELECT department_id FROM EMPL_TEST WHERE last_name =
'Mourgos');
```